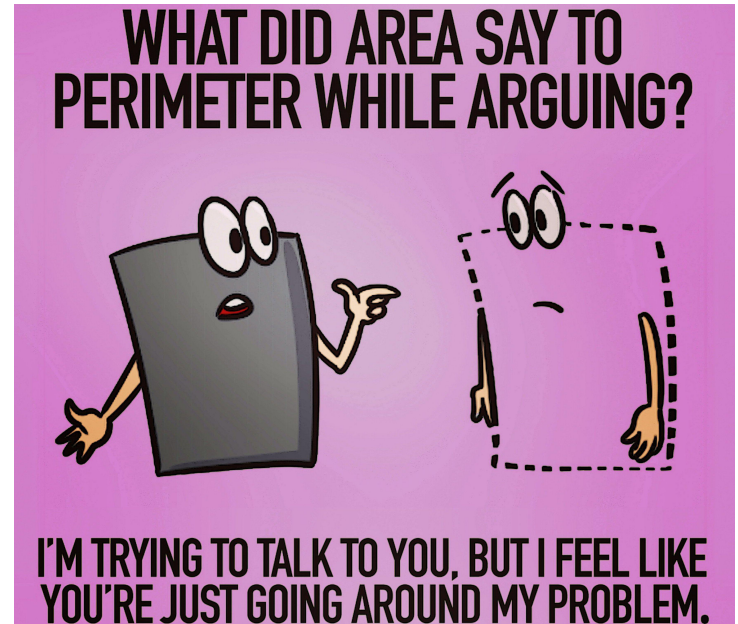


Help your contractor design your new kitchen space. First order of business, choose room dimensions. Make sure you make it clear to the contractor that while technically speaking a room size of 1ft x 100ft has the required area of 100ft² ... you will not be building it. (No room should have a 1ft width.)



Input

You will receive a rectangle's area as an integer on a single line.

100

Output

Output all possible integer side lengths the rectangle could have to generate that area, printing from largest to smallest. Do not repeat a combination of sides.

E.g. do not list both "50 x 2" and "2 x 50", as that is redundant. All side lengths must be larger than 1.

50 x 2
25 x 4
20 x 5
10 x 10

Discussion

Reminder: have you run your solution against **all** of the student data sets?

Additional Examples

Input 2	Output 2	Input 3	Output 3
1024	512 x 2 256 x 4 128 x 8 64 x 16 32 x 32	8	4 x 2